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Exploring the medicinal potential of *Amaranthus spinosus* leaves used by the Santal tribe for boil treatment: A qualitative phytochemical study

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Abstract- Medicinal plants are essential in the traditional healthcare practices of numerous indigenous communities, including the Santal tribe in India. *Amaranthus* leaves are prominently used in ethnomedicine for treating skin boils and associated inflammatory conditions. Despite their extensive conventional use, empirical evidence of therapeutic efficacy remains limited. This study aims to investigate the qualitative phytochemical composition of *Amaranthus* leaves, providing scientific insights into their medicinal properties. Fresh leaves frequently employed in Santal healing rituals were collected, washed, shade-dried, and ground into powder. Methanolic extracts were prepared and analyzed using standard qualitative phytochemical analysis according to established methodologies described by Harborne, Trease and Evans, and Kokate. The analysis identified a variety of bioactive compounds, such as alkaloids, flavonoids, tannins, glycosides, terpenoids, carbohydrates and phenolic constituents. These metabolites are extensively linked to antimicrobial, anti-inflammatory, analgesic, and wound-healing properties, providing scientific validation for the ethnomedicinal use of *Amaranthus* leaves in the treatment of boils. Flavonoids and phenolics are recognized for their potent antioxidant and anti-inflammatory characteristics, whereas tannins play a role in antimicrobial activity and tissue regeneration. The results indicate that the Santal tribe's traditional knowledge corresponds with biochemical evidence, thereby substantiating the therapeutic potential of *Amaranthus* leaves. This study represents a preliminary step towards the pharmacological assessment and potential advancement of botanical treatments for dermal infections. It emphasizes the significance of safeguarding the indigenous knowledge system and amalgamating it with contemporary scientific methodologies.

Keywords: *Amaranthus spinosus*, Santal Tribe, Boil, Qualitative Phytochemical Analysis, Flavonoids.

INTRODUCTION

Medicinal plants have been integral to traditional healthcare systems for centuries, especially in developing and rural communities with limited access to modern medical facilities. According to the World Health Organization, nearly 80% of the global population relies on plant-based remedies for primary healthcare needs.[1] India is acknowledged as a megadiverse country and possesses a rich historical tradition of plant-based medicine, exemplified by systems such as Ayurveda, Siddha, and Unani.[2]

Alongside classical systems, tribal communities nationwide maintain extensive ethnomedicinal knowledge, which is preserved through oral traditions.[3] The Santal tribe, a prominent Indigenous group in eastern India, possesses significant traditional ecological knowledge, especially concerning medicinal plants used for treating infections, wounds, and inflammatory conditions.[4] Research on Santal ethnomedicine has identified various plant species used to treat skin conditions such as boils, cuts, and abscesses.[5]

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Rapid cultural and environmental changes have jeopardized the intergenerational of this knowledge.[6] Boils, also known as furuncles, are acute skin infections primarily caused by *Staphylococcus aureus*, resulting in painful inflammation and the formation of pus.[7] In tribal and rural areas with restricted access to contemporary healthcare, the use of herbal treatments persists. In the Santal community, crushed leaves of *Amaranthus* species are commonly applied topically to boils to mitigate swelling and infection[8]. *Amaranthus* species exhibit a high concentration of secondary metabolites, including flavonoids, phenolics, tannins, terpenoids, saponins and alkaloids.[9] Phytochemicals exhibit antimicrobial, anti-inflammatory, antioxidant, and wound-healing properties, potentially validating the plant's traditional use in treating skin infections.[10] Phytochemical screening is recognized as a crucial initial step in identifying bioactive components in medicinal plants.[11,12] Ethno pharmacological research validates traditional medicinal knowledge, supports the discovery of novel therapeutic compounds, and promotes the preservation of Indigenous practices.[13] Examining the phytochemical profile of *Amaranthus* leaves utilised for boil treatment by the Santal tribe integrates traditional knowledge with contemporary scientific insights. This study examined the qualitative phytochemical screening of *Amaranthus* leaves, which are traditionally utilized by the Santal tribe. The study seeks to identify major phytochemical groups to provide evidence for the plant's ethnomedicinal role and establish a basis for future pharmacological research.

MATERIALS & METHODS

Study area and ethnobotanical information

Santal tribal communities living in the Laludih villages of Deoghar district provided ethnobotanical information about the traditional use of *Amaranthus* leaves to treat boils. Informal conversations with elderly community members who apply *Amaranthus* leaves topically to infected skin areas, as well as traditional healers, provided preliminary information. All respondents provided their prior informed consent, and the ethics of ethnobotanical research were also adhered.

Collection of Plant Materials

During the growing season, fresh, healthy *Amaranthus spinosus* L. leaves were gathered from the surrounding area in the early morning. Members of the Santal tribe, who frequently use the leaves to treat boils and related skin infections, provided traditional ethnomedical knowledge that informed the choice of this plant species. Informal conversations with knowledgeable local practitioners were used to record information about the medicinal use.

Preparation of Plant Extract

The freshly harvested leaves were carefully rinsed with distilled water to guarantee total removal of surface contaminants after being thoroughly cleaned under running tap water to remove adhering dust particles, soil, and other unnecessary debris. After cleaning, the leaf samples were evenly spread out on fresh blotting sheets and left to dry for fifteen days at room temperature (22–25°C) in the shade until a constant weight was reached, signifying that all remaining moisture had been removed. To reduce the loss of heat-labile phytoconstituents and avoid deterioration from direct sunlight exposure, shade drying was chosen. A clean, dry electric grinder was used to grind the dried leaves into a fine, uniform powder. The powdered substance was promptly put into sealed containers and kept there. The powdered material was promptly transferred to airtight containers and kept dry to avoid moisture absorption, oxidative changes, and photodegradation until additional extraction and phytochemical analyses were carried out.

Solvent Extraction

In a clean conical flask, 2 g of the dried leaf powder were precisely weighed and macerated with 40 mL of hydroalcoholic solvent (methanol: distilled water) for hydroalcoholic extraction. To ensure effective extraction of phytochemical constituents, the mixture was shaken occasionally and kept at room temperature for a full day. Following the extraction phase, a clear filtrate was obtained by first filtering the mixture through muslin cloth and then using Whatman No. 1 filter paper. To maximize the recovery of bioactive compounds, the residue was extracted again using a new solvent. The filtrates were then combined and concentrated by solvent evaporation at room temperature. The resultant crude hydroalcoholic extract was then put into sterile, appropriately labelled containers and kept at 4°C until additional qualitative and quantitative phytochemical analyses were performed in accordance with standard laboratory procedures.

Qualitative Phytochemical Screening

Standard phytochemical procedures were employed for the qualitative detection of major classes of secondary metabolites using well-established and recognized protocols.[11,12,14] The following tests were performed:

1. Test for Alkaloids

Wagner's reagent was used to confirm the presence of alkaloids by observing the formation of reddish-brown precipitate or coloration.

2. Test for Flavonoids

The alkaline reagent test was performed. Development of an intense yellow color, which becomes colorless on addition of dilute HCl, indicated the presence of flavonoids.

3. Test for Tannins

Braymer's test was conducted, where a blue or greenish coloration confirmed the presence of tannins.

4. Test for Saponins

The foam test was performed by shaking the extract vigorously with distilled water; persistent foaming for 10-15 minutes indicated saponins.

5. Test for Terpenoids

Salkowski's test was used, where the immediate formation of a reddish-brown precipitate confirmed the presence of terpenoids.

6. Test for Glycosides

Keller–Kelliani's test was performed for cardiac glycosides; a reddish-brown ring indicated a positive reaction.

7. Test for Phenolic Compounds

The ferric chloride test was used to confirm phenolics by the appearance of a deep blue or black color.

8. Test for Carbohydrates

Molisch's reagent, along with 1 ml of concentrated sulfuric acid, was added down the side of the test tube. Then allow the mixture to stand for 2-3 minutes. Observe for the formation of red or dull violet color at the interface of the two layers, which is a positive result.

9. Test for Quinones

Test with 0.5ml of conc. Hydrochloric acid and observed for the formation of a yellow precipitate or color change.

10. Test for Resins

Distilled water was added, and turbidity was observed.

11. Test for Coumarins

The formation of a yellow color after adding 10% NaOH indicated the presence of coumarins.

Qualitative Data Interpretation

The appearance of noticeable color changes, precipitate formation, or turbidity development seen during routine phytochemical tests was used to interpret the qualitative phytochemical results. To verify the existence or lack of phytochemical groups, each reaction was closely inspected and cross-checked with recognized qualitative indicators. To improve the reliability of observation, consistency of visual responses across multiple tests was taken into consideration. To clearly illustrate the qualitative phytochemical profile of *Amaranthus spinosus* leaf extracts, the collected results were methodically recorded and displayed in tabular form. The Santal tribe's traditional use of the leaves to treat boils and related skin infections was scientifically validated by the identification of major classes of bioactive constituents made possible by this interpretative approach.

Safety and Ethical Considerations

Standard biosafety precautions, such as the appropriate use of personal protective equipment and the safe handling and disposal of chemicals and plant materials in compliance with institutional guidelines, were followed throughout all laboratory procedures. The ethical guidelines for research involving indigenous communities were followed when collecting ethnobotanical data. All local participants provided informed consent after being fully informed about the study's goals, and the informants' anonymity and confidentiality were upheld. The information gathered was used only for academic and scientific purposes, and the study respected the Santal community's traditional knowledge and cultural values of the Santal community.

RESULTS

Qualitative Phytochemical Screening

Qualitative analysis of the methanolic extract of *Amaranthus* leaves revealed the presence of several major groups of bioactive compounds, indicating its rich phytochemical composition. Preliminary screening confirmed the presence of important secondary metabolites, such as alkaloids, carbohydrates, flavonoids, tannins, saponins, terpenoids, glycosides, and phenols. These phytoconstituents are known for their therapeutic potential, particularly in treating skin-related disorders. The results are summarized below:

The presence of these phytochemicals suggests that *Amaranthus* leaves possess significant medicinal potential, particularly in terms of antimicrobial and wound-healing activity, which aligns with their traditional use by the Santal tribe for treating boils.

Table 1. Qualitative phytochemical analysis of *Amaranthus spinosus* leaf extract

Sl. No.	Phytochemical constituent	Qualitative test	Observation	Result
1	Alkaloids	Wagner's test	Reddish-brown color	+
2	Carbohydrate	Molisch's test	Dull violet color at the interface	+
3	Flavonoids	Alkaline reagent test	Yellow coloration	+
4	Quinones	Hydrochloric acid test	No change	-
5	Tannins	Braymer's test	Greenish coloration	+
6	Saponins	Froth test	Stable froth for few minutes	+
7	Terpenoids	Salkowski test	Reddish-brown ring at the interface	+
8	Glycosides	Keller–Killiani test	Reddish-brown ring at the junction	+
9	Phenolic compounds	Ferric chloride test	Green coloration	+
10	Resins	Water test	No turbidity	-
11	Coumarins	Sodium hydroxide test	No change	-

Note: (+) Presence of phytochemical constituent; (–) Absence.

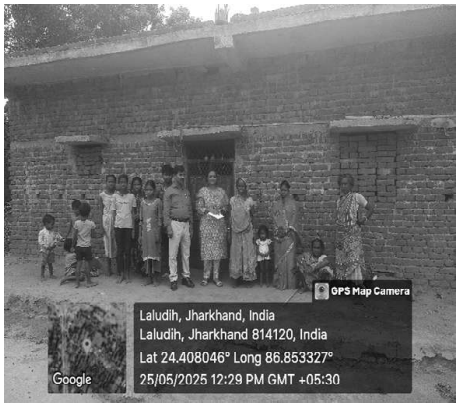


Fig. 1- Field Survey



Fig. 2- Field Survey



Fig. 3- *Amaranthus spinosus* plant



Fig. 4- *Amaranthus spinosus* dry leaves



Fig. 5- Plant extract

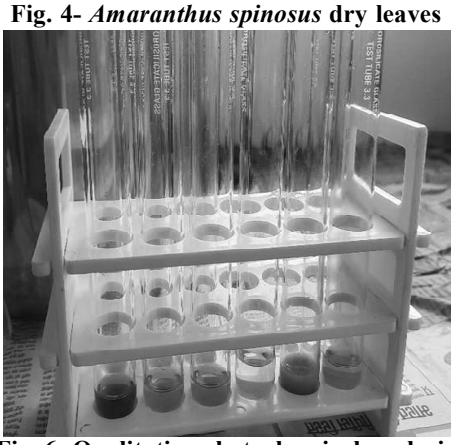


Fig. 6- Qualitative phytochemical analysis

DISCUSSION

1. Ethnomedicinal Relevance- Freshly crushed *Amaranthus* leaves are traditionally applied topically by the Santal tribe to treat localized infections and skin boils. The presence of bioactive substances involved in antimicrobial, anti-inflammatory, and wound healing pathways is frequently correlated with ethnomedical uses.[14] According to earlier research, *Amaranthus* species have strong phytochemicals that can treat skin infections.[15]

2. Presence of Alkaloids and Terpenoids- Because alkaloids and terpenoids are metabolites with antimicrobial, analgesic, and anti-inflammatory qualities, their detection is important.[16] These substances may support the traditional therapeutic methods of Santal healers by reducing infections and inflammation associated with boils.

3 Flavonoids and Phenolic Compounds- The leaf extract contained significant amounts of phenolics and flavonoids. These groups are well known for their antioxidant activity, which is crucial for wound healing and tissue repair.[11] Additionally, flavonoids possess antibacterial and anti-inflammatory properties that support their ethnomedical use in treating infected skin lesions.[17]

4. Glycosides and Saponins- Because saponins have surface-active antimicrobial qualities and are frequently associated with the disruption of bacterial cell membranes, their presence is significant.[18] Although cardiac glycosides are better known for their ability to modulate the heart, they also have cytotoxic and antimicrobial properties that are important for controlling the pathogenic microorganisms that cause skin infections.

5. Tannins- A slight presence of tannins, which have antimicrobial and astringent qualities.[12] They aid in the reduction of exudates and the contraction of wounds.

6. Comparison with earlier research- The findings of the present study reveal a phytochemical profile that closely aligns with previously reported investigations on various *Amaranthus* species. Flavonoids, phenolics, and terpenoids were found in *Amaranthus* leaf extracts, highlighting the effectiveness of these compounds in wound healing.[19] In methanolic extracts of *Amaranthus spinosus*, strong antimicrobial activity was discovered that correlated with tannins, saponins, and flavonoids.[20] Leafy vegetables with high saponin and tannin content add to their antimicrobial and detoxifying qualities.[21] Together, these studies validate the ethnopharmacological practice of treating boils with *Amaranthus* leaves and support the findings reported here.

SUMMARY OF FINDINGS

Several significant secondary metabolites with known medicinal value were found in the leaves of *Amaranthus spinosus,* according to the current qualitative phytochemical analysis. Since alkaloids, tannins, and saponins have been shown to inhibit pathogenic microorganisms and promote wound healing, their detection suggests strong antimicrobial potential.[16,22] This discovery offers scientific support for the Santal tribe's custom of using *Amaranthus* leaves to treat boils, which are frequently linked to bacterial infections.

Given that terpenoids have been shown to modify inflammatory pathways and reduce tissue inflammation, their presence suggests significant anti-inflammatory qualities.[11,20] This kind of exercise could help reduce boil-related pain, swelling, and redness. Furthermore, the qualitative identification of phenolic compounds and flavonoids emphasizes the plant material's antioxidant potential. These substances have been shown to scavenge free radicals, shield cellular components from oxidative damage, and promote tissue regeneration and repair.[16,23]

Overall, the phytochemical profile found in this study provides a scientific foundation for the traditional use of *Amaranthus spinosus* leaves to treat skin conditions and strongly supports their ethnomedicinal significance. The results support the plant's potential as a natural source for the creation of plant-based antimicrobial and anti-inflammatory agents by indicating that the plant's therapeutic efficacy may be the consequence of the synergistic action of several bioactive compounds.

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